

Q. List types of authentication. Explain any one.

Types of Authentication:

1. **Passwords**
2. **Fixed Password**
3. **One-Time Password (OTP)**
4. **Biometric Authentication** (fingerprint, face recognition)
5. **Token-based Authentication** (smart cards, security tokens)

One-Time Password (OTP)

- OTP is a **secure authentication method** in which a unique password is generated for each login or transaction
- It is sent to the user via **SMS, email, or authentication apps**
- OTP is **valid only for a short time** and can be used only once
- After use, it becomes **invalid and cannot be reused**
- It provides **better security** than fixed passwords as it reduces the risk of hacking

Working of OTP:

1. User enters username and password
2. System generates a unique OTP
3. OTP is sent to registered mobile/email
4. User enters the OTP
5. If OTP is correct, access is granted

Advantages:

- Prevents unauthorized access
- Protects against password theft
- Enhances security in online transactions

OTP Authentication Diagram

